

TrES-3b

R-2529

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110501 · created 2026-08-25T22:59:37+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455682.79899 +/- 0.00088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.019
Transit depth (Rp/Rs)^2	3.73 +/- 0.74 [%]
Orbital Inclination (inc)	81.61 +/- 0.52
Airmass coefficient 1 (a1)	1.0185 +/- 0.01
Airmass coefficient 2 (a2)	-0.014 +/- 0.0041
Scatter in the residuals of the lightcurve fit is	0.97 %
Best Comparison Star	#2 - [117, 265]
Optimal Aperture	3.7
Optimal Annulus	9.16
Transit Duration (day)	0.0559 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.019 vs 0.16309 (deviation 1.55σ)
Duration fit vs published	0.0559 d vs 0.0567 d (deviation 0.14σ)
Mid-time O–C	-3.0 min
Depth SNR	10.0
Residual scatter	0.97 %

Light curve

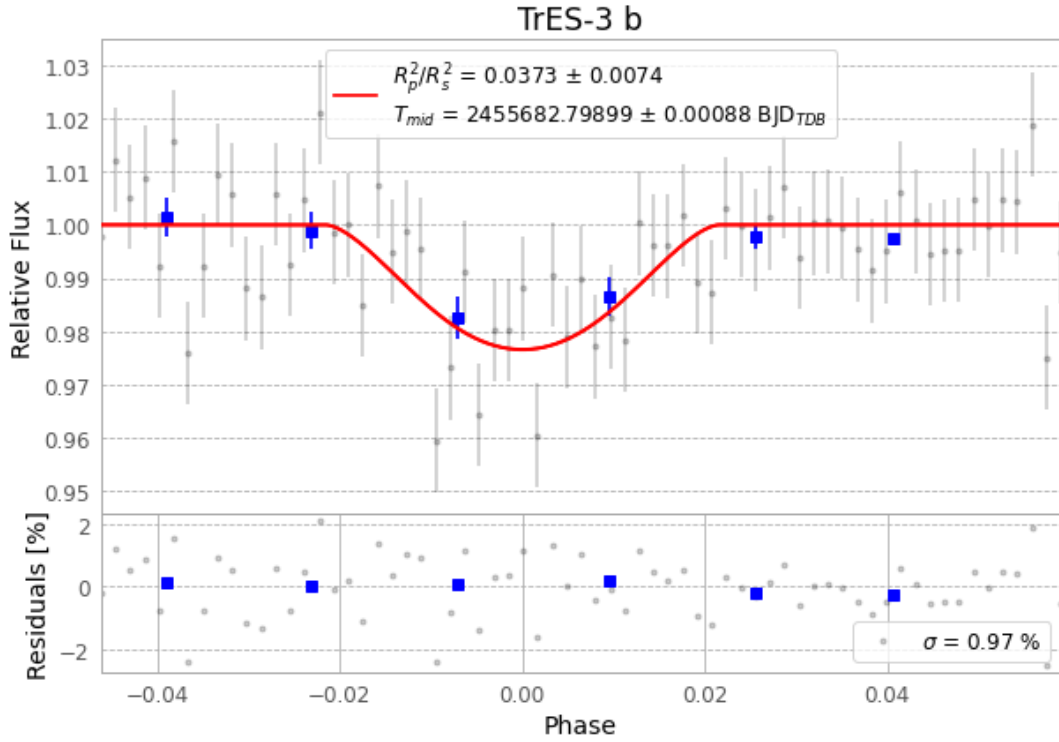


Figure 1 — FinalLightCurve_TrES-3 b_2011-04-30.png (SHA-256 489cc8c6f2c4ef18...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [189, 183], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2fd019c2c2288b49...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-3110501053912.FITS → TRES-3110501085713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110501.log (0e86d1801317...), FinalLightCurve_TrES-3 b_2011-04-30.png (489cc8c6f2c4...), FinalLightCurve_TrES-3 b_2011-04-30.csv (f113a01dfac3...)

Compute resources

Wall time 102.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 22:59Z.